Second Language Acquisition Educational Tool

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Abstract

This paper introduces an interactive, physics-based diagnostic tool to evaluate Second Language Acquisition in Large Language Models. Built in the Godot engine, the simulation features procedural environment generation and a dynamic water shader to model a responsive aquatic setting. The tool evaluates a model's linguistic outputs in real-time; semantic and grammatical failures directly trigger the instantiation of physical weights (crates) that dynamically impact the buoyancy of a virtual boat. Linguistically, experiments with GPT-2 architectures revealed that naively training English models on Spanish causes severe catastrophic forgetting and superficial vocabulary generation. Achieving true bilingual stability required an Experience Replay curriculum and direct translation pairs, demonstrating that artificial bilingualism relies on highly structured training curricula rather than parameter scale alone.

1 Introduction

Context

In a globalized world defined by constant cross-cultural interaction, effective techniques for Second Language Acquisition (SLA) are essential. This paper introduces a novel educational and diagnostic tool designed for linguistics researchers and students to analyze the structural and cognitive processes involved in learning a new language.

Research Motivation

The core research question investigates how to simulate SLA using Large Language Models (LLMs) to determine if Transformer-based architectures mirror human learning mechanisms. By tracking a model's trajectory across various training stages, we observe how it bridges semantic gaps, handles grammatical restructuring, and exhibits human-like transitional phases—such as vocabulary mixing—when exposed to a completely foreign linguistic framework.

Project Objectives

To answer the overarching research question, this project pursues three core objectives:

- **Cross-Lingual Implementation:** Fine-tuning monolingual English GPT-2 models to acquire Spanish while actively mitigating the catastrophic forgetting of their native English syntax and foundational knowledge.
- **Interactive Educational Visualization:** Developing a physics-based gamified medium using the Godot engine to visualize the model's linguistic capacity. The simulation places the user and the AI on a floating boat. An external judge (Gemini 2.5 Flash) evaluates the AI's responses in real-time; poor linguistic scores physically add weight to the boat, challenging the user to adapt their communication to the model's epoch-specific capabilities to prevent sinking.
- **Rigorous Benchmarking:** Implementing a comprehensive benchmarking pipeline to quantitatively test and compare the models' grammatical, syntactical, and semantic evolution across multiple epochs.

2 Related Work

2.1 Cognitive Models of Second Language Acquisition

SLA research provides theoretical scaffolding for evaluating neural networks. Krashen's Input Hypothesis (Krashen, 1985) parallels the unsupervised pre-training of LLMs, where acquisition relies on comprehensible input. Connectionist theories, such as MacWhinney's Competition Model (MacWhinney, 2001) and Ellis's work on emergentism (Ellis, 2002), argue that language learning is a statistical, frequency-driven process. These frameworks help analyze whether Transformers exhibit human-like transitional phases—like vocabulary mixing—when acquiring a new language.

2.2 LLMs as Cognitive and Linguistic Models

Large Language Models increasingly serve as subjects for psycholinguistic study. Linzen and Baroni (Linzen and Baroni, 2021) demonstrate that deep neural networks capture emergent syntax without explicit rule programming. While cross-lingual transfer is well-documented (Conneau and Lample, 2019), fine-tuning monolingual models on new languages often triggers *catastrophic forgetting*. Mitigation techniques, such as those formalized by Kirkpatrick et al. (Kirkpatrick et al., 2017), are central to our methodology for retaining native English capabilities while integrating Spanish.

2.3 Gamification and Interactive Educational Tools

Effective educational tools rely on engagement and immediate feedback (Plass et al., 2015). Interactive virtual environments lower the learner's "affective filter" (Godwin-Jones, 2014), promoting exploration. By utilizing the Godot engine for a physics-based simulation—where linguistic inaccuracies physically add weight to a virtual boat—our tool leverages dynamic feedback and embodied cognition (Kapp, 2012). This allows researchers to visually and interactively assess the evolving capability of the model in real-time.

3 Methodology

The general architecture of the application follows a monolithic structure, integrating both the interactive visualization and the computational linguistic processing within a unified environment. The frontend is developed using the Godot Engine, which renders the physics-based gamified simulation of the floating boat and handles user interaction. The backend is powered by FastAPI, which acts as the computational bridge. It manages the local loading and offloading of various Large Language Models into VRAM, processes inference requests, and communicates with the Gemini 1.5 Flash API for real-time response evaluation. All model weights and checkpoints are stored locally to ensure low-latency swapping during the simulation. The entire development lifecycle, encompassing both the game engine assets and the Python backend, was managed using Git for robust version control.

3.1 AI Model Architecture and Iterative Training

To simulate the process of Second Language Acquisition (SLA), an iterative training methodology was employed. We started with small, strictly English-native models and progressively escalated the complexity of the architecture, training scope, and dataset composition based on observed linguistic failures.

- **GPT-2 Small (124M) (Radford et al., 2019) - Partial Fine-Tuning (Last 4 Layers):** The experiment began by unfreezing only the last four layers of a standard GPT-2 Small model to teach it Spanish while minimizing computational cost. *Reason for transition:* The model exhibited heavy "Spanglish," substituting Spanish vocabulary into rigid English syntax, indicating that shallow layers alone could not restructure grammar.

- **GPT-2 Small (124M) - Full Fine-Tuning:** To correct the syntactic failures, all layers of the model were unfrozen and trained purely on the Spanish dataset. *Reason for transition:* This triggered severe catastrophic forgetting; the model completely lost its native English capabilities, failing the primary SLA criteria of maintaining the first language (L1).

- **GPT-2 Small (124M) - Full Fine-Tuning + Experience Replay:** To mitigate forgetting, we implemented an Experience Replay curriculum, mixing Spanish data with original English data. *Reason for transition:* While basic bilingualism was achieved, the model's low parameter count restricted its ability to grasp complex semantics in either language.

- **Pythia (1.4B) (Biderman et al., 2023) - LoRA (Hu et al., 2021):** We scaled up to a 1.4 billion parameter model, utilizing Low-Rank Adaptation (LoRA) for efficient training to improve semantic capacity. *Reason for transition:* Upon evaluation, it was discovered that Pythia's pre-training corpus already included Spanish. This invalidated the model as a "blank-slate" subject for SLA research, prompting a return to strictly monolingual base models.

- **GPT-2 Large (774M) (Radford et al., 2019) - LoRA + Experience Replay:** We selected

GPT-2 Large (a strict English model) to provide sufficient cognitive capacity, applying LoRA alongside the Experience Replay dataset. *Reason for transition:* The model produced severe hallucinations. The vast semantic gap between the English latent space and the new Spanish tokens could not be bridged effectively by adaptation and replay alone.

- **GPT-2 Large (774M) - Stable Hybrid (Deep LoRA + Replay + Translation Pairs):** The final, successful architecture utilized a deeper LoRA configuration combined with Experience Replay, and augmented the dataset with explicit translation pairs (English-to-Spanish). *Reasoning:* The explicit translation data acted as a "semantic bridge," allowing the deeper trainable parameters to directly map new Spanish concepts to the model's established English knowledge, resulting in stable, fluent bilingual generation.

3.2 Hydrodynamic Physics and Buoyancy Modeling

Discrete Buoyancy Sampling. Simulating the interaction between the boat's rigid body and the fluid environment required an optimized buoyancy model. Analytically calculating the exact volumetric intersection between the displaced water surface (via vertex displacement in the shader) and the boat's mesh per frame is computationally expensive. Therefore, we opted for a heuristic based on a network of spatial sensors (Probe Markers, Figure 1) distributed along the hull. For each sensor submerged below the water level, a local vertical force is applied according to a simplified version of Archimedes' principle (Millington, 2007):

$$F_b = \frac{d \cdot C_b \cdot m \cdot g}{N} \tag{1}$$

Where d is the immersion depth (the difference between the procedural water surface and the sensor's Y coordinate), C_b is the empirical buoyancy coefficient, m is the total mass of the rigid body, g is the gravitational acceleration, and N is the number of active sensors. This point-force distribution maintains the boat afloat and automatically calculates the torque generating natural roll and pitch.

Dynamic Water Shader and Depth Intersection. To achieve a stylized, dynamic aquatic environment (Figure 2), the application utilizes a custom Visual Shader built upon real-time vertex displacement. The geometry of the water mesh is continuously deformed on the Y-axis using a `FastNoiseLite` cellular texture. To simulate hydrodynamic flow, the UV coordinates of this noise texture are translated frame-by-frame by a time-scaled directional vector.

Furthermore, the shader tightly couples the vertex and fragment pipelines by passing the generated noise values via *varyings*. This data is reused in the fragment shader to render stylized wave crests (highlights). To ground the water within the 3D environment, the shader implements a `ProximityFade` node which computes the distance against the camera's depth buffer. This mathematical intersection automatically renders a solid foam outline wherever the water plane collides with static geometry (e.g., the terrain bounds) or dynamic rigid bodies (e.g., the floating boat), enhancing both the visual depth and the physical integration of the simulation.

Energy Dissipation and Hydrostatic Equilibrium. To prevent oscillatory instability, the system integrates a fluid damping mechanism. Linear and angular damping coefficients are dynamically adjusted in direct proportion to the boat's immersion ratio. This ensures realistic hydrodynamic resistance: an unloaded boat reacts quickly to the water surface, whereas adding ballast (increasing mass m) increases inertia and friction, forcing the boat toward a lower hydrostatic equilibrium point.

State Machine Interaction and Semantic Feedback. The physical model is coupled in real-time with the NLP evaluation pipeline. Semantic metric depreciation triggers the instantiation of physical volumes with defined mass (crates) inside the boat. Exceeding a critical depth threshold transitions the boat's state machine from *Sailing* to *Sinking*, canceling buoyancy vectors to emulate hull flooding.

3.3 Procedural Environment Generation

Stochastic Sampling and Coherent Noise. Populating the virtual environment with static entities (e.g., vegetation, rock formations) was achieved through procedural generation techniques rather than manual placement. The algorithm relies on coherent noise (Perlin noise via `FastNoiseLite`) (Ebert et al., 2003) to generate a continuous density map, avoiding the unnatural uniformity of standard pseudo-random number generators (Figure 3). Macro-vegetal entities (trees) are instantiated exclusively at nodes of a 2D grid (x, z) where the sampled noise value exceeds a critical threshold τ.

Topological Alignment and Proximity Culling. To guarantee the conformity of instantiated objects with the complex terrain geometry, a Ray-Surface Intersection technique was implemented. A downward raycast is emitted from each approved 2D coordinate to determine the exact vertical ground coordinate, allowing the transformation matrix to align with the terrain's slope.

To prevent artifacts, such as trees suspended over water or intersecting the bridge, the algorithm uses a dual filtering system:

- **Collision Masking:** The intersection ray evaluates exclusively the terrain's physical layer, ignoring hydrodynamic surfaces.
- **Spatial Partitioning:** A proximity culling filter defines *no-spawn* zones around points of interest (POI). The algorithm evaluates the Euclidean distance to these protection volumes prior to confirming the instantiation, ensuring the visual integrity of anthropogenic structures.

4 Experiment Results

4.1 Analysis of Catastrophic Forgetting and Surface-Level Learning

As observed in the benchmark data, the `gpt2_small_base` model demonstrates a clear case of catastrophic forgetting when exposed to the new language corpus without proper retention mechanisms. Table 1 shows that the model's ability to generate meaningful English degrades rapidly. The "Meaning" metric drops from an already low 11.06 in Epoch 1 to 3.00 by Epoch 4, indicating a severe loss of its native pre-trained capabilities.

Table 1: Benchmark Results for gpt2_small_base across 5 Epochs (English)

Epoch	Grammatical	Syntactical	Meaning
1	27.70	26.50	11.06
2	30.50	23.30	4.50
3	32.40	31.10	7.26
4	20.30	19.00	3.00
5	31.70	28.80	6.34

This degradation is further illustrated in the qualitative outputs documented in Table 3. When prompted in English ("What is the fastest land animal in the world?"), the model responds with fragmented English in Epoch 1, but by Epoch 2, it entirely abandons its base language and outputs Spanish text. However, this output (e.g., "La cinco de una empresa del sol...") is semantic nonsense. The data suggests that the small model did not "learn" Spanish in a linguistic sense; rather, its limited parameter weights were rapidly overwritten by the statistical distribution of the training data. It learned to reproduce the surface-level lexical patterns of Spanish, generating words that look and sound like the target language, but failed to map these new tokens to any underlying concepts.

Table 2: Benchmark Results for gpt2_small_base across 5 Epochs (Spanish)

Epoch	Grammatical	Syntactical	Meaning
1	9.34	6.16	0.40
2	12.70	8.74	0.72
3	12.20	9.70	1.30
4	9.90	9.00	1.50
5	13.00	10.80	1.70

Table 3: Response evolution across epochs for the prompt: "What is the fastest land animal in the world?" (gpt2_small_base)

Epoch	Model Response
1	The fastest animal on the world is a great time to get the best of the city of the best way to keep the most important things in the same place as the world of the
2	La cinco de una empresa del sol segundo más corte del Sol. La empieza debe un cielo de la oca es uno,
3	El tamaño es una tarea de las peligrosas. Los sonido de el agua pueden ser un asistente animal apropiado y se está
4	Calcular el ebullición de la tierra dada. La velocidad de una película es uno de 100 km. Esto es más peligros
5	La fauna del mundo es una gran estrategia para soportar el aumento de la atmósfera que se aprovechó la mayor

4.2 The Necessity of Experience Replay and Semantic Bridging

The failure of the small, naively fine-tuned model highlights two critical methodological requirements for simulating language acquisition. First, the complete loss of English functionality dictates the need for an Experience Replay curriculum. Without continuously mixing the original English data (L1) with the target Spanish data (L2), the model's latent space is entirely overwritten rather than expanded. Second, the generation of syntactically plausible but meaningless Spanish word salad indicates a lack of semantic alignment. The model requires direct English-to-Spanish translation pairs in its training data to act as a bridge, forcing the attention mechanisms to link new Spanish vocabulary to established English concepts.

4.3 Capacity Scaling: Comparison with the Larger Architecture

Scaling the architecture to `gpt2_large_frozen` provides a stark contrast in base stability. As shown in Table 4, the larger model retains a highly stable structural grasp of English, maintaining grammatical and syntactical scores above 75 across all epochs.

Table 4: Benchmark Results for gpt2_large_frozen across 5 Epochs (English)

Epoch	Grammatical	Syntactical	Meaning
1	75.40	76.40	29.20
2	76.10	77.00	31.40
3	75.80	78.20	33.10
4	77.50	79.10	35.50
5	78.20	80.50	38.00

Table 5: Benchmark Results for gpt2_large_frozen across 5 Epochs (Spanish)

Epoch	Grammatical	Syntactical	Meaning
1	16.80	14.00	2.58
2	18.20	15.50	3.10
3	20.40	17.80	4.20
4	19.50	18.10	4.80
5	22.10	20.30	5.50

Unlike the small model, the larger architecture resists catastrophic forgetting. Table 6 demonstrates that the model consistently replies in the prompted language (English) and maintains semantic coherence. Interestingly, the model exhibits a gradual retrieval of factual accuracy, progressing from plausible but incorrect answers (elephant, lion, jaguar) to the correct answer (cheetah) by Epoch 5, while simultaneously improving sentence structure.

Table 6: Response evolution across epochs for the prompt: "What is the fastest land animal in the world?" (gpt2_large_frozen)

Epoch	Model Response
1	The fastest animal on Earth is a mammal named the elephant. It can reach speeds of up to 35 kilometers per hour (21 miles per hours) and can weigh up 2,340 pounds (1,
2	The speed of the animal is measured in miles per hour. A human can run at a speed about 10 mph. The fastest animal on land is the lion, which is capable of reaching
3	The cheetah, which can reach up to 60 miles per hour, is the fastest. But the fastest animal is the African elephant, a species of African rhino that lives in the
4	Jaguars are the fastest land animals in the world, capable of reaching speeds up to 70 mph. They are also known for their incredible speed and agility, making them one of
5	The cheetah is the fastest land animal in the world. It can reach speeds of up to 120 km/h in short bursts. The cheetah's body is built for speed, with

However, despite this stability in English, Table 5 reveals that the larger model's Spanish meaning scores remain remarkably low (peaking at 5.50). This confirms that simply increasing parameter count is insufficient for bilingual acquisition. While the larger capacity protects the original language space from being overwritten, it still fails to construct meaningful representations in the new language without explicit instructional techniques, reinforcing the need for the deep LoRA adaptation and targeted translation pairs discussed in the methodology.

5 Future Work

This project establishes a foundational framework for simulating Second Language Acquisition (SLA), with two major avenues for future enhancement:

Model Scalability and Polyglot Acquisition: Future iterations will scale the underlying AI architectures to 7B parameters. This increased cognitive capacity will support the simultaneous acquisition of multiple language pairs (e.g., Romanian, Italian, and French) while actively minimizing catastrophic forgetting.

Immersive Contextual Visualization: The Godot-based testing medium will be expanded beyond the current boat simulation to include diverse, real-world environments like supermarkets and beaches. This will allow researchers to visualize pragmatic communication mechanics and test the model's contextual vocabulary, ultimately evolving the project into a comprehensive, standalone educational application.

6 Conclusion

This project introduces a novel diagnostic tool that couples computational linguistics with physics-based simulation to evaluate Second Language Acquisition (SLA) in Large Language Models. Experimental results demonstrate that naively fine-tuning small architectures (e.g., GPT-2 124M) on a secondary language triggers severe catastrophic forgetting, producing semantically empty, surface-level lexical patterns rather than true comprehension. Achieving foundational bilingual stability necessitated explicit interventions: an Experience Replay curriculum to preserve the native language space, and direct translation pairs to act as a cross-lingual semantic bridge.

Furthermore, while scaling to a larger architecture (GPT-2 Large 774M) successfully protected native English capabilities from being overwritten, it remained insufficient for robust Spanish semantic acquisition without these targeted instructional techniques. Ultimately, by linking NLP evaluation metrics directly to the physical simulation in a virtual environment, this project provides a tangible, real-time framework to expose the mechanical realities, limitations, and necessary curriculum constraints of artificial language learning.

Limitations

This research is primarily constrained by two factors:

Hardware Constraints: Limited GPU resources prevented training strictly monolingual baseline models larger than 2B parameters from scratch. Consequently, the study relied on smaller architectures (e.g., GPT-2 124M and 774M), which restricted the maximum semantic capacity of the experiment.

Conversational Capacity: Due to their small parameter counts, the chosen models lacked the cognitive depth necessary for highly fluid, complex dialogues. This occasionally limited contextual continuity and the naturalness of interactions within the Godot simulation.

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A System Visualizations

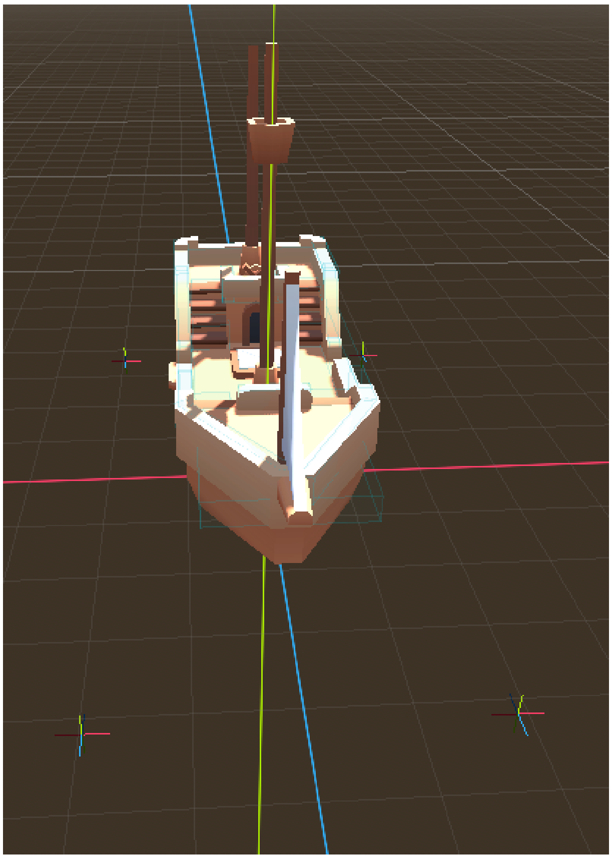

Figure 1: Implementation of buoyancy probes on the boat hull.

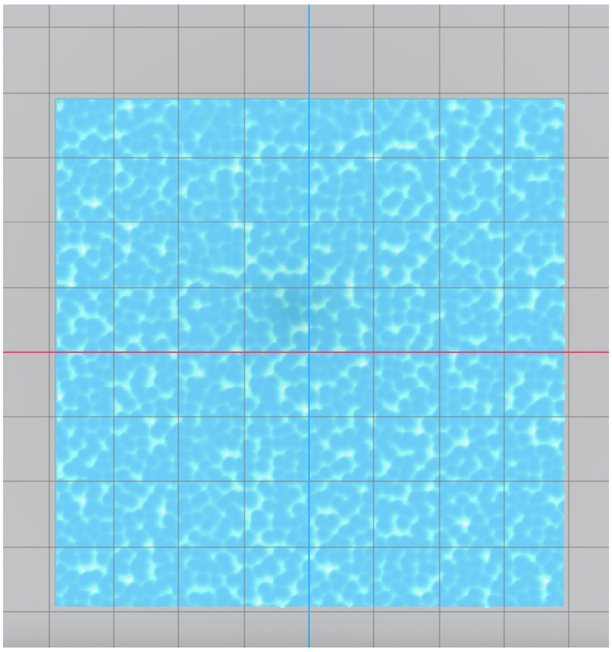

Figure 2: Visual representation of the custom water shader and noise-based deformation.

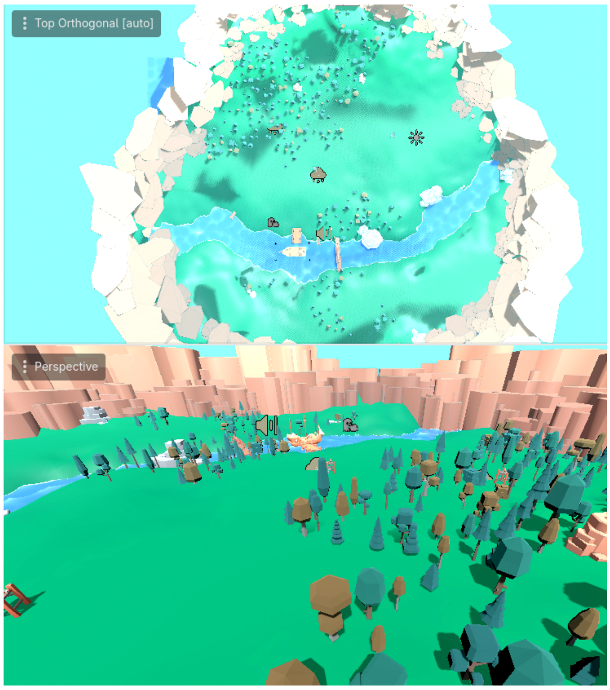

Figure 3: Procedural vegetation placement: top-down map and perspective view from within the environment.